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BANGALORE · INDIA

Develop and validate a synthetic dementia dataset through train test swap

Samiksha Sandeep Zokande (2448354)
Saikat Maity (2448353)
Ravichandra D (2448349)

Guided By - Dr. Jayapriya J

Mission
Christ University is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment

Vision
Excellence and Service

Core Values
Faith in God | Moral Uprightness
Love of Fellow Beings | Social Responsibility
Pursuit of Excellence

Context

- Problem statement
- Model design
- Numerical
- Text
- Image
- Challenges
- Future Work
- References

Problem Statement

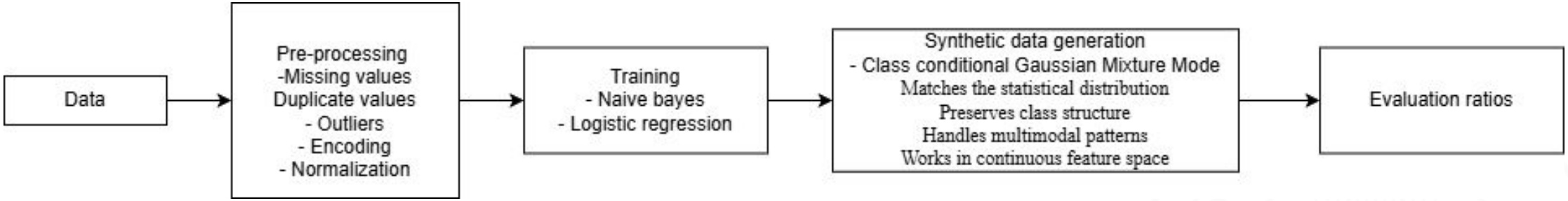
Problem Statement:

- Develop and validate a synthetic dementia dataset
- Systematically comparing real vs. synthetic data
- Through
  - Train–test swaps
  - 25% real-data evaluation
  - mixing-ratio ablations (0–100%)
  - cross-site validation

Output:

Synthetic data with a robust accuracy and indication of good dataset so that it can be used in any form of research or project.

Numerical


Data
Pre-processing
-Missing values
Duplicate values
- Outliers
- Encoding
- Normalization
Training
- Naive bayes
- Logistic regression
Synthetic data generation
- Class conditional Gaussian Mixture Mode
Matches the statistical distribution
Preserves class structure
Handles multimodal patterns
Works in continuous feature space
Evaluation ratios

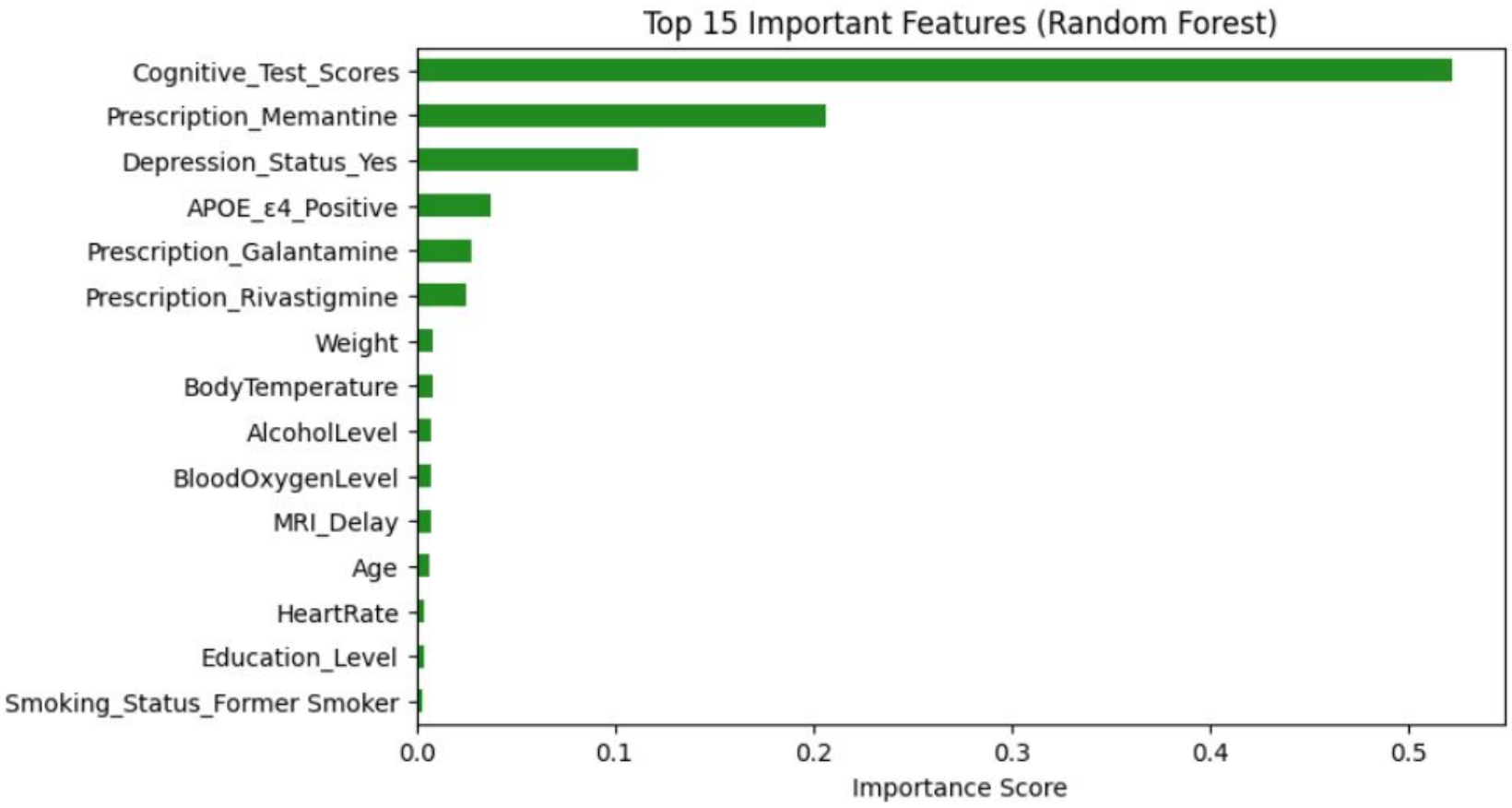
Numerical

Features

```
Data columns (total 24 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Diabetic                   1000 non-null   int64
 1   AlcoholLevel               1000 non-null   float64
 2   HeartRate                  1000 non-null   int64
 3   BloodOxygenLevel           1000 non-null   float64
 4   BodyTemperature            1000 non-null   float64
 5   Weight                     1000 non-null   float64
 6   MRI_Delay                  1000 non-null   float64
 7   Prescription               485 non-null    object
 8   Dosage in mg               485 non-null    float64
 9   Age                        1000 non-null   int64
 10  Education_Level            1000 non-null   object
 11  Dominant_Hand              1000 non-null   object
 12  Gender                     1000 non-null   object
 13  Family_History             1000 non-null   object
 14  Smoking_Status             1000 non-null   object
 15  APOE_ε4                    1000 non-null   object
 16  Physical_Activity          1000 non-null   object
 17  Depression_Status          1000 non-null   object
 18  Cognitive_Test_Scores      1000 non-null   int64
 19  Medication_History         1000 non-null   object
 20  Nutrition_Diet             1000 non-null   object
 21  Sleep_Quality              1000 non-null   object
 22  Chronic_Health_Conditions  821 non-null    object
 23  Dementia                   1000 non-null   int64
```

Numerical

Features



Numerical

Models

Naive Bayes & Random Forest

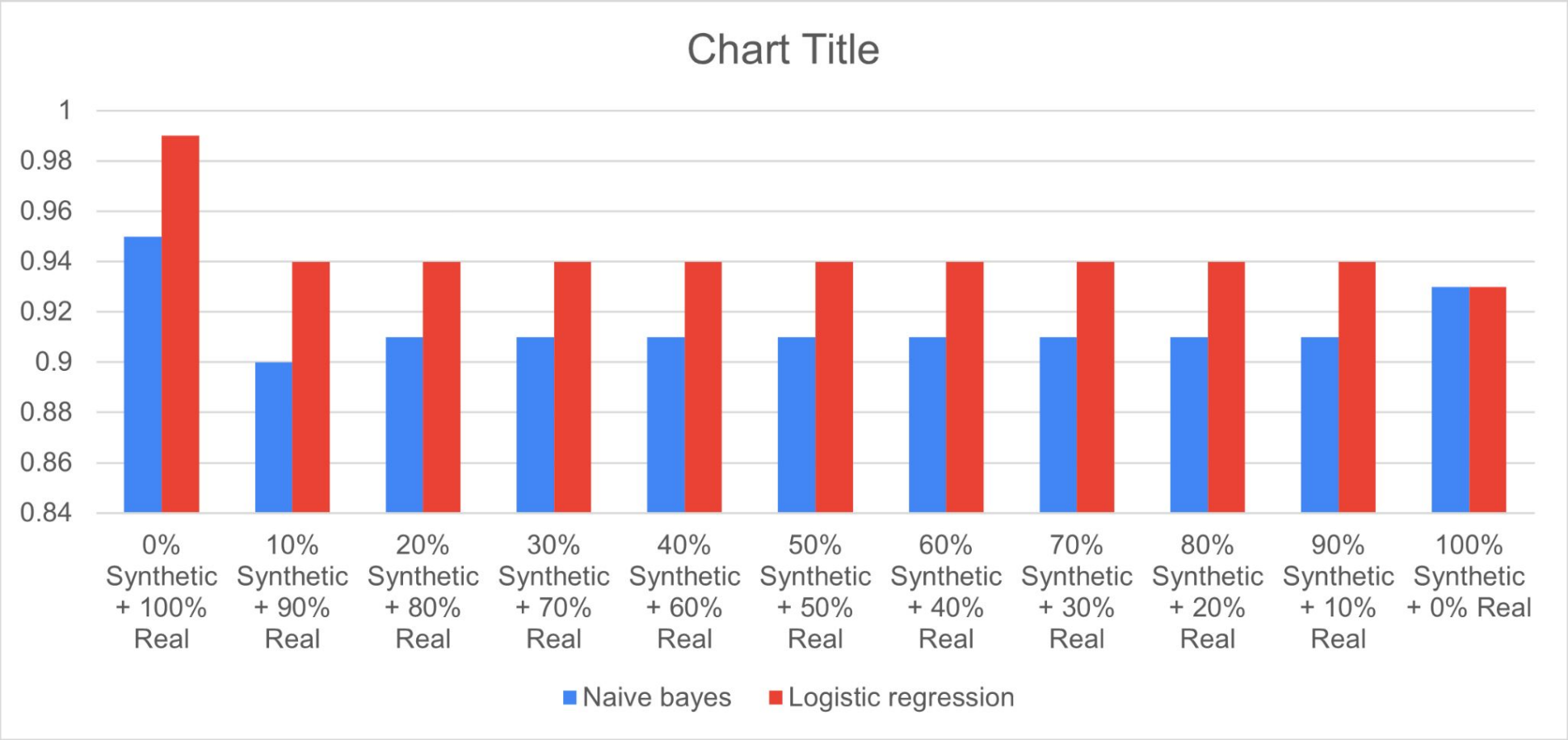
- Limited samples
- Noisy measurements
- High variance risk

Class Conditional Gaussian Mixture Model

- Trains reliably with limited data
- GMMs are specifically designed to model continuous-valued data
- Each Gaussian component captures a sub-population of patients
- No label leakage
- No overlap-induced mislabeling

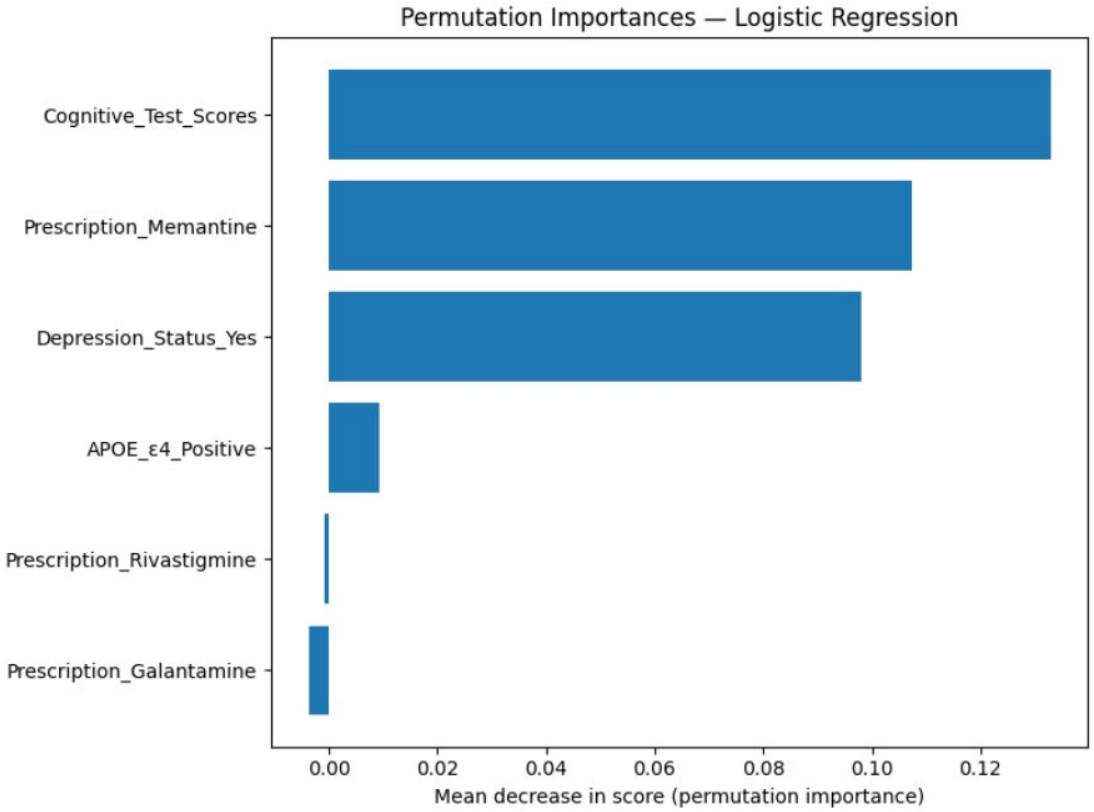
Numerical

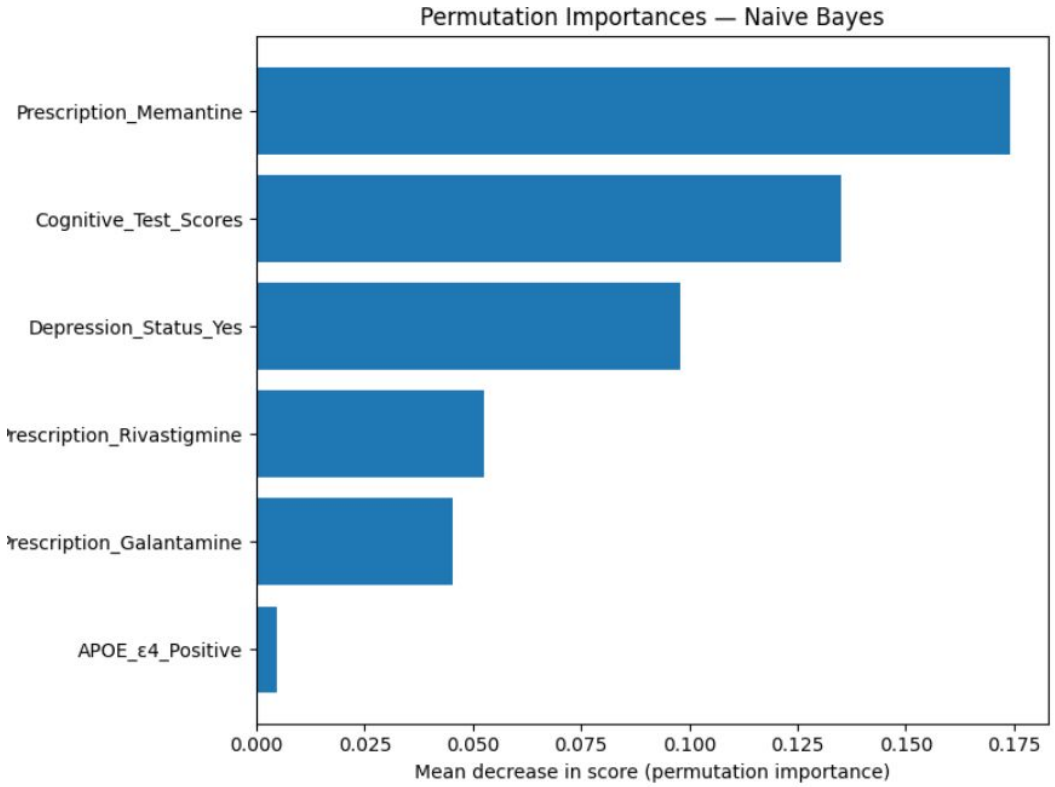
Results



Numerical

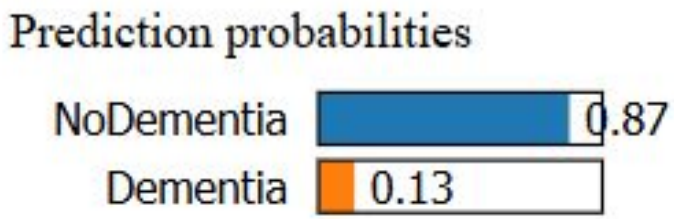
Results :- XAI

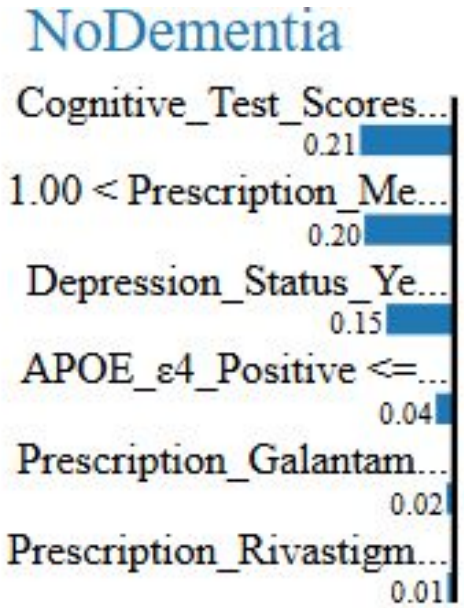




Numerical

Results :- XAI





Numerical

Challenges

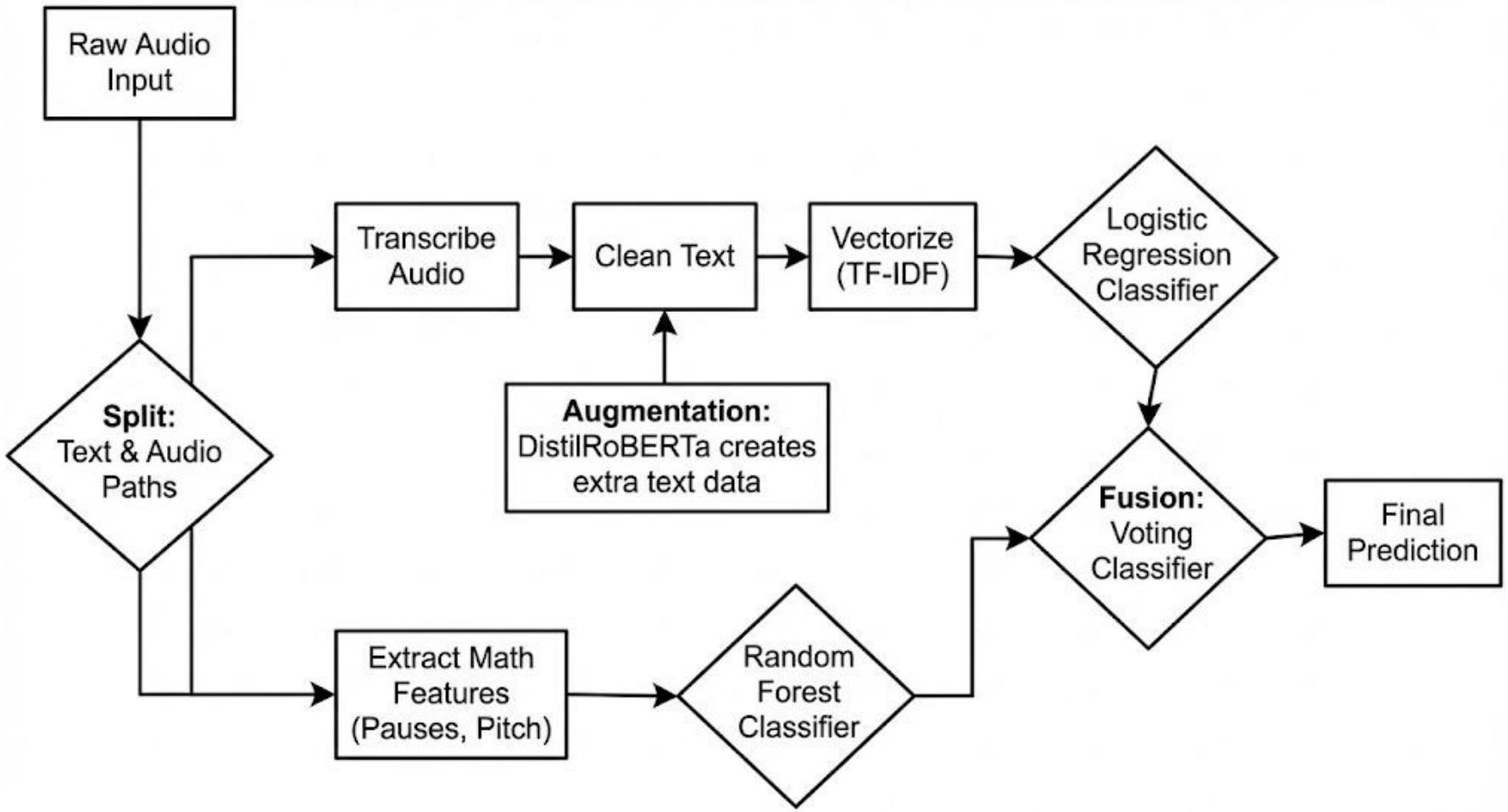
Numerical

1. Limited availability of real-world dementia datasets for model training and benchmarking.
2. The dataset was quite small.
3. Different data schema

Audio Data

- Dataset: The project utilized the **DementiaBank Pitt Corpus**, a comprehensive database of audio recordings and transcripts from individuals with dementia and healthy controls.
- Initial Size: The raw dataset consisted of **454 audio samples (Total: 454 transcripts)**.
- Class Distribution:
  - Original Data: Highly imbalanced, with approximately **72% 'No Dementia'** (Healthy) and 28% 'Dementia' samples.
  - Augmented Data: To address this imbalance, the dataset was expanded using synthetic data generation, resulting in a new total of **1,054 samples (454 original + 600 synthetic)**.
  - Final Balance: The training set was balanced to ensure fair model learning, with roughly equal representation of both classes.

Audio Data Methodology



Audio MODEL

Model 1: Logistic Regression (Text Baseline)

- **Why used:** A strong, interpretable baseline for text classification. It works well with high-dimensional sparse data like TF-IDF vectors.
- **Role:** Analyzed *linguistic* features (vocabulary loss).
- **Configuration:** `class_weight='balanced'` was used to handle the initial class imbalance.

Model 2: Random Forest (Audio Baseline)

- **Why used:** An ensemble method that handles non-linear relationships well. Perfect for numerical *acoustic* features (pitch, energy, pauses).
- **Role:** Analyzed *paralinguistic* features (how it was said).

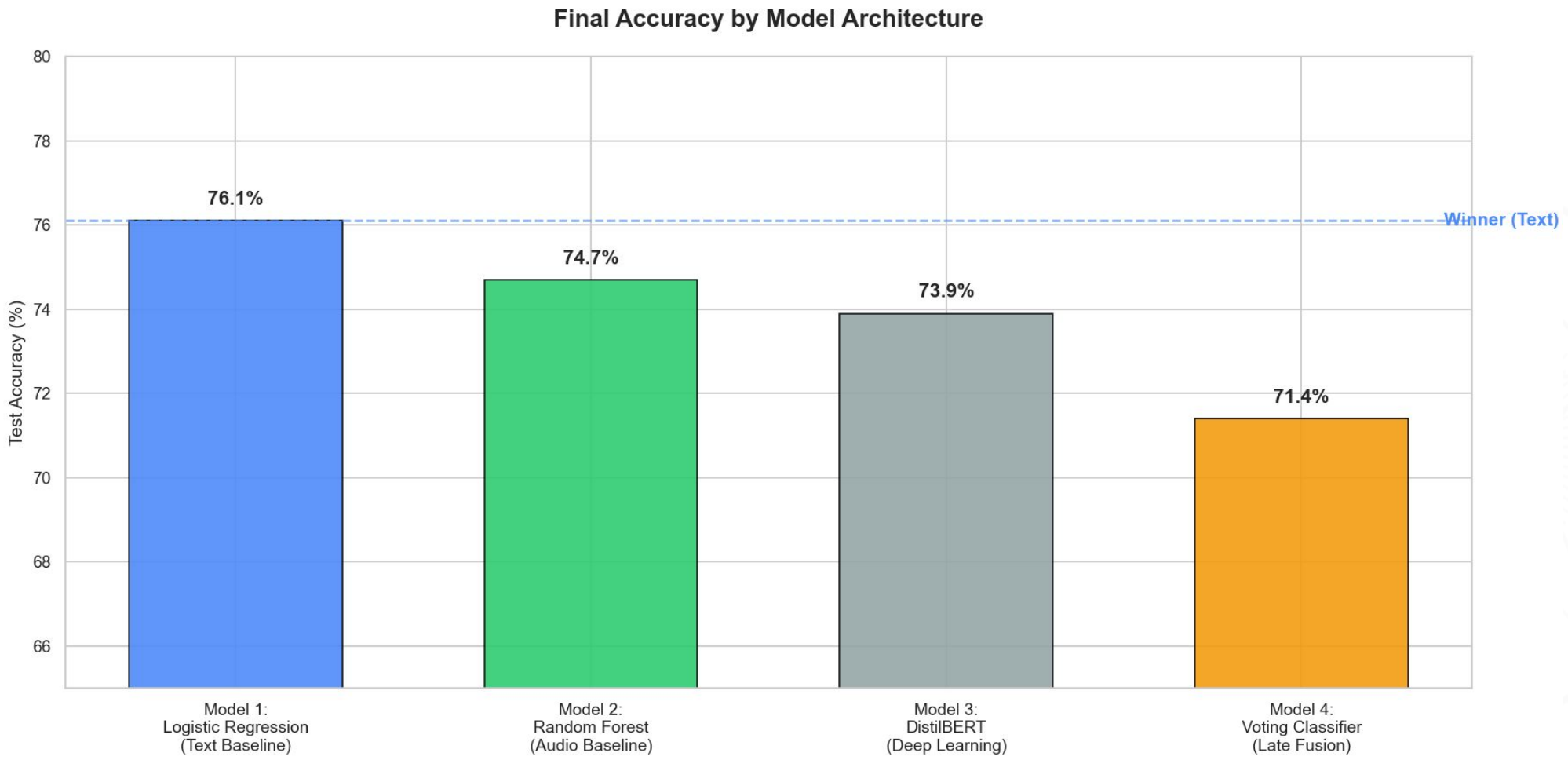
Model 3: DistilBERT (Deep Learning)

- **Why used:** A state-of-the-art Transformer model pre-trained on massive text data.
- **Role:** Attempted to capture deep semantic context and nuance.
- **Outcome:** Proved to be "data-hungry" and overfitted on the small dataset, performing worse than simpler models.

Model 4: Voting Classifier (Late Fusion)

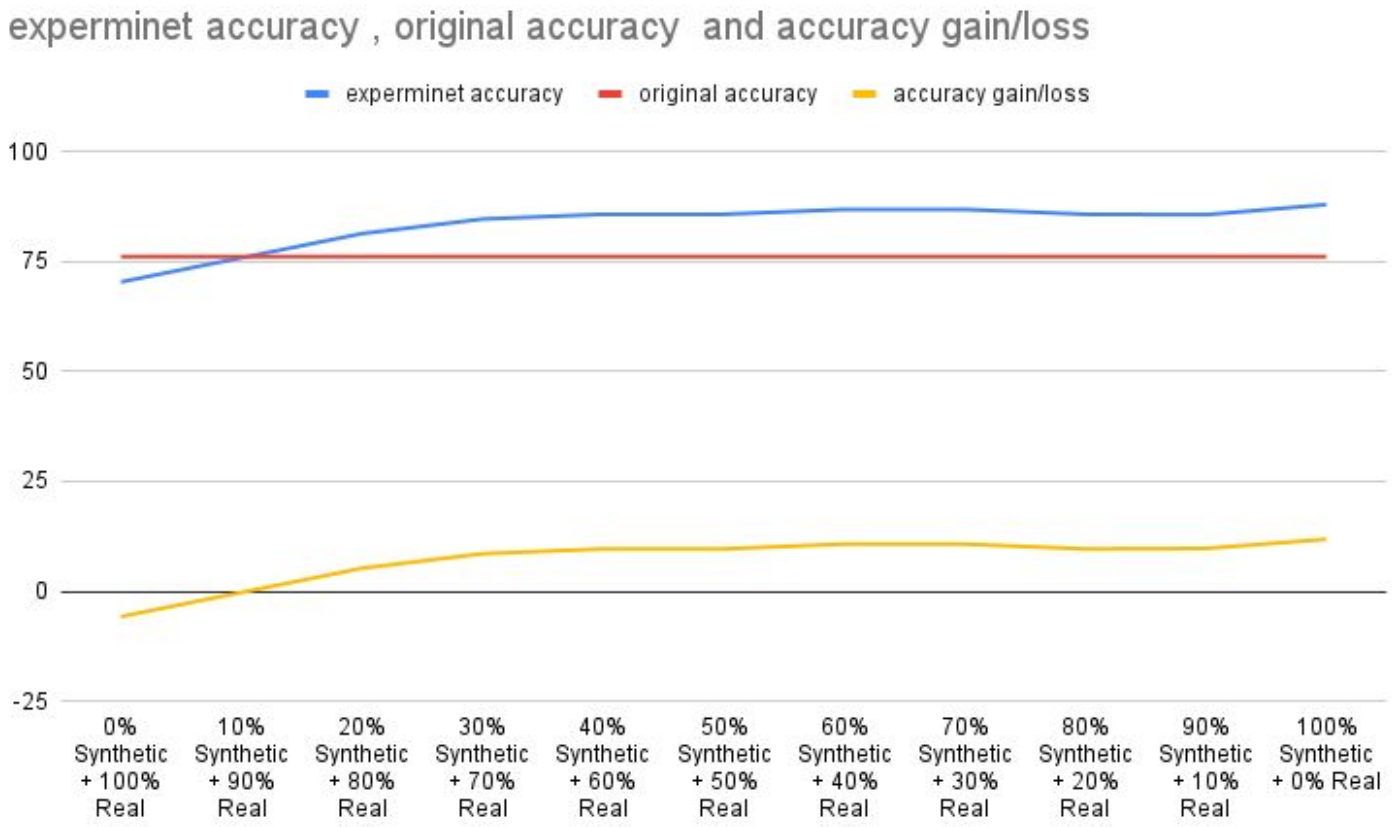
- **Why used:** To combine the strengths of both Text and Audio models.
- **Method:** Trained Text and Audio models separately, then averaged their predictions (weighted towards Text).
- **Outcome:** Improved over early fusion but didn't beat the text-only baseline, proving text is the dominant signal.

Model Accuracy



Mixed Data

Results



XAI

Linguistic Markers (Feature Importance):

- **Visual:** A bar chart showing the top words contributing to a "Dementia" diagnosis vs. "Healthy."
- **Findings:** The model flagged words like **"um," "uh,"** and generic terms (fillers) as strong indicators of dementia, aligning with clinical symptoms of **anomia** (word-finding difficulty).

Acoustic Visualization:

- **Visual:** The **Spectrogram** (heatmap of frequency) and **Waveform** (loudness over time).
- **Explanation:**
  - **Healthy Waveform:** Continuous, smooth speech patterns.
  - **Dementia Waveform:** Fragmented signal with frequent flat lines (silence), visually proving the **high pause rate** detected by the model.

Clinical Simulation (What-If Analysis):

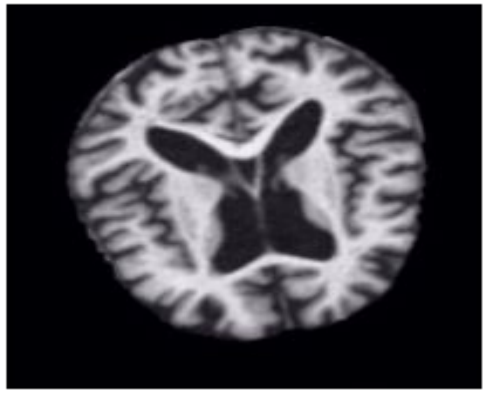
- **Tool:** The "Manual Input" tab in your app.
- **Action:** Demonstrated how changing a single variable (e.g., lowering Speech Rate from 2.5 to 1.5) drastically increases the predicted risk score, confirming the model has learned valid medical thresholds.

Dataset

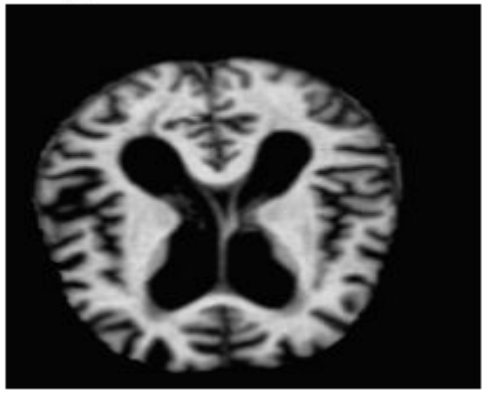
The Dementia MRI Disease dataset is a valuable resource for researchers and health medicine applications. This dataset focuses on the classification of Dementia disease based on MRI scans it contains 5120 images.

The dataset consists of brain MRI images labeled into 4 categories:

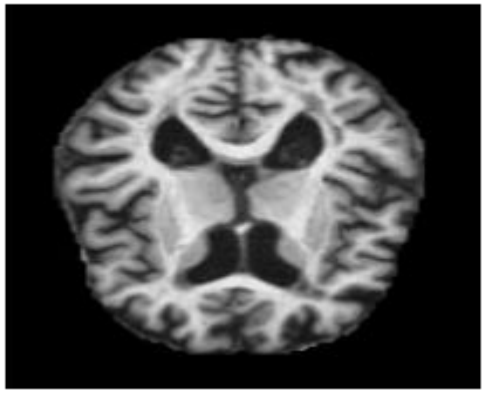
ModerateDemented1000



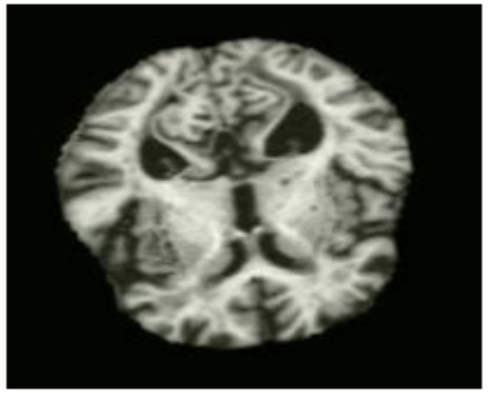
VeryMildDemented1000



NonDemented1000



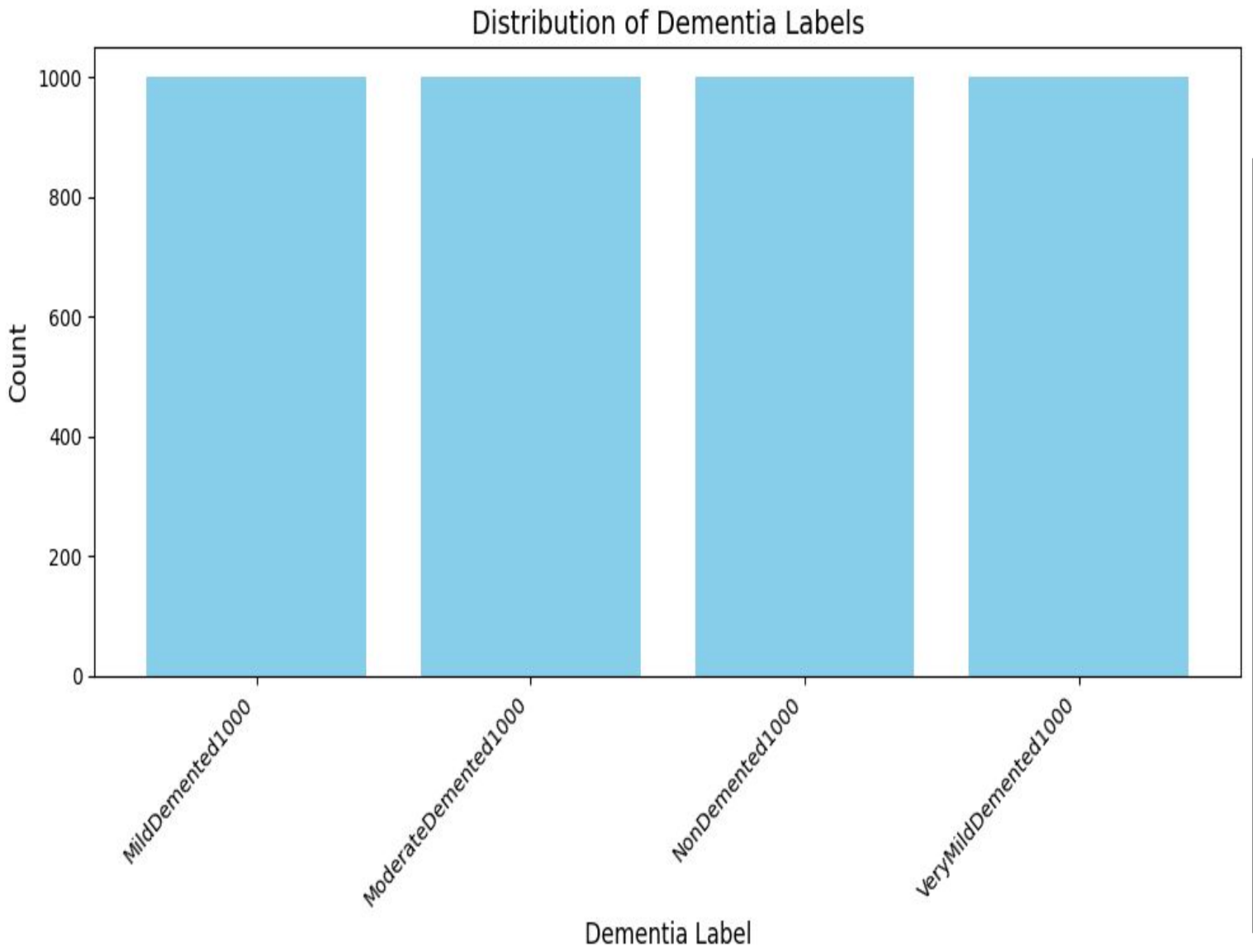
MildDemented1000



Image

Image Dataset

- Import Image Dataset
- Data Preprocessing
  - Resizing
  - Grayscale/RGB Conversion
  - Gaussian Blur
  - Data Balancing
  - Classification of Image
- Data Augmentation
  - Reduce Contrast
  - Horizontal vs. vertical clips
  - Zoom , crop, rotate
- Feature extraction
  - ResNet50
  - VGG16
  - Efficient NET

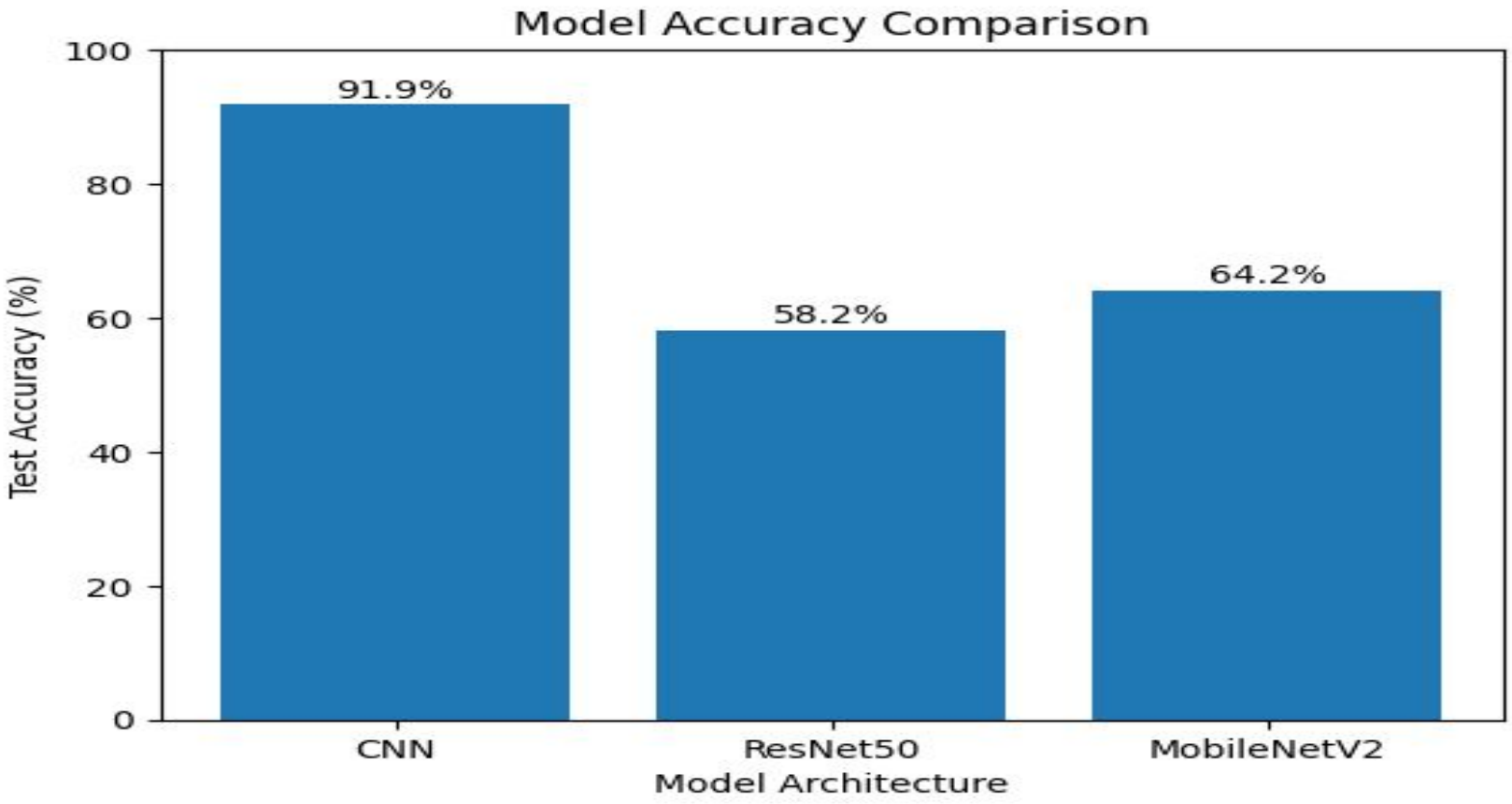


Models

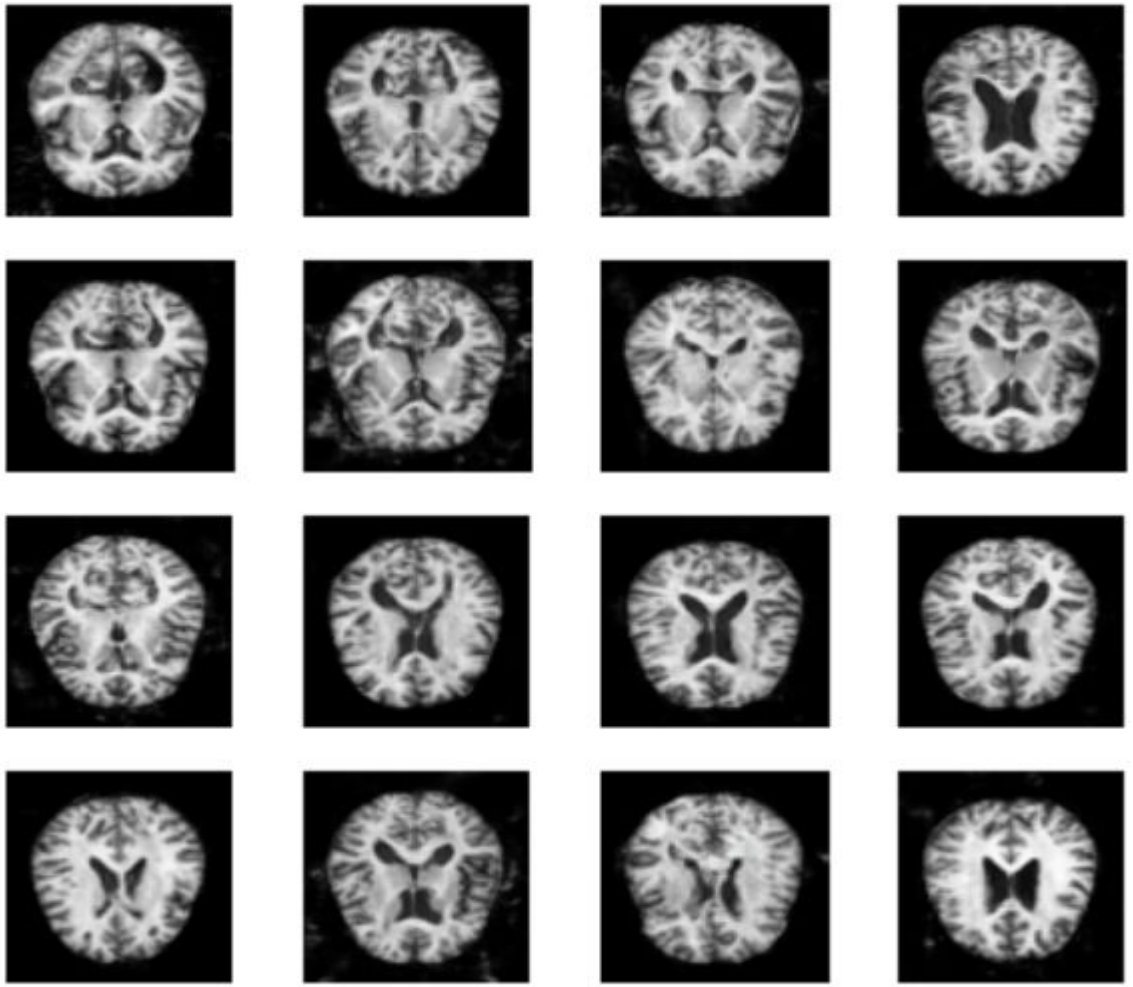
- CNN - grayscale, multiple convolutional blocks
- Mobilenetv2 - speed and efficiency
- Resnet50 - uses skip connections,solve gradient degradation problem
- GAN
- VAE

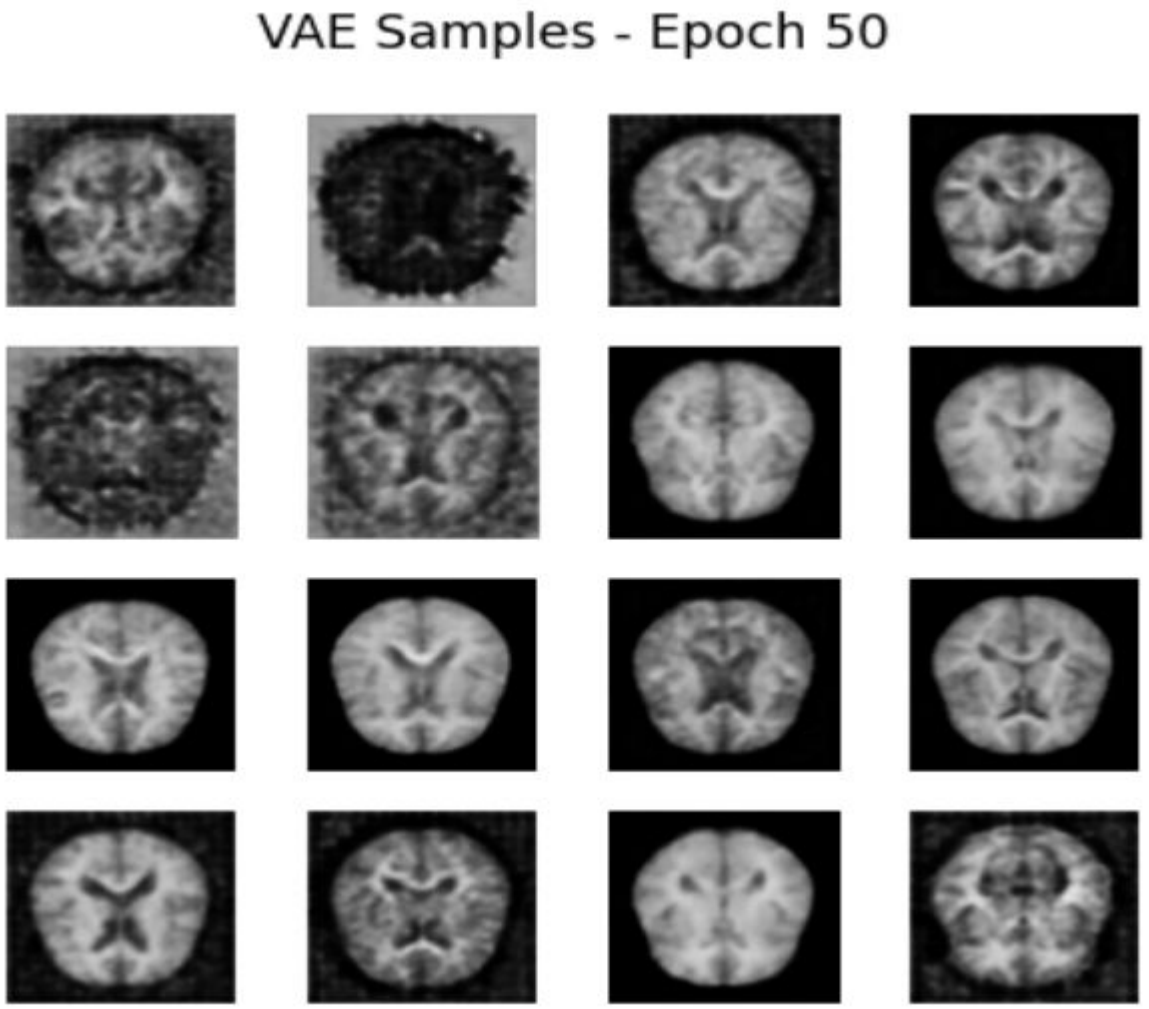
Image

Results

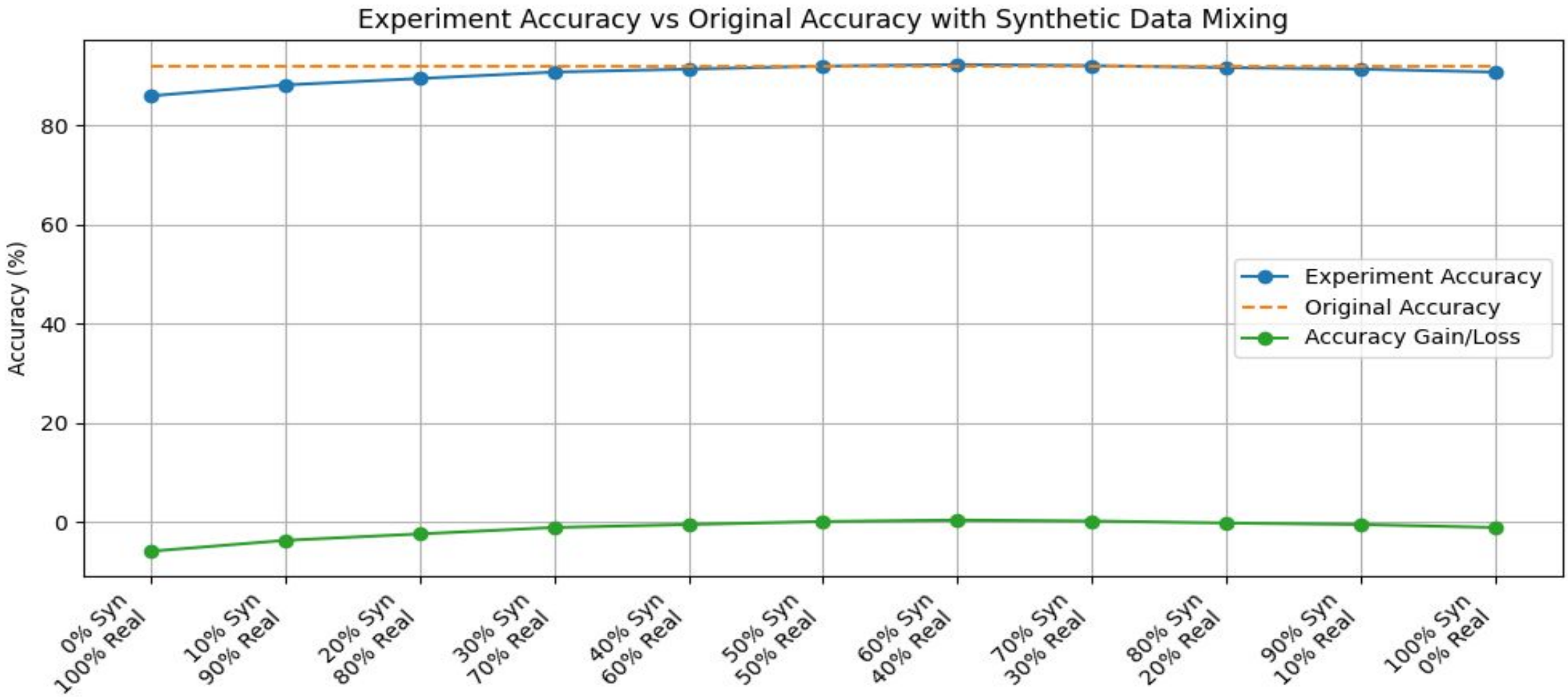


Synthetic Images of GAN and VAE





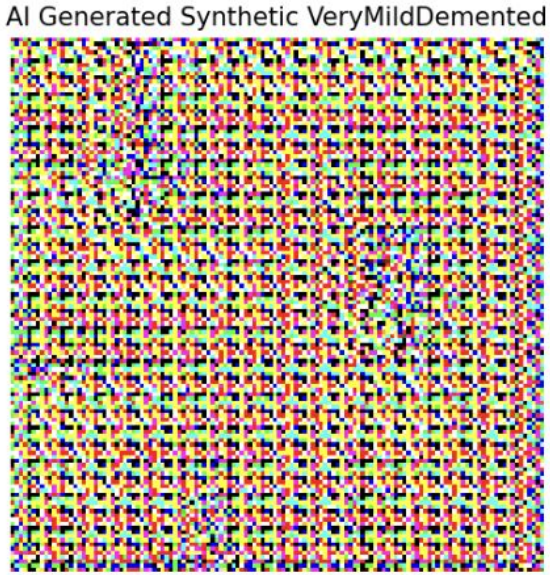
Mixed ratio

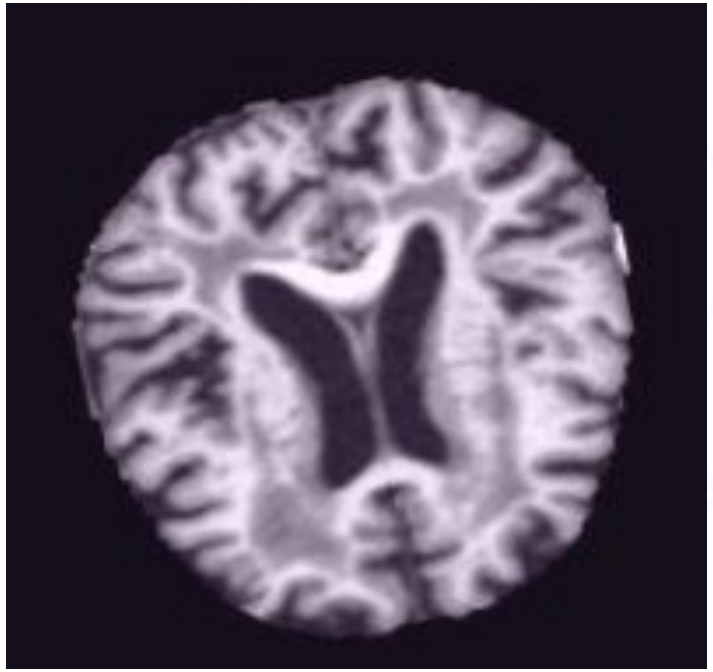

Experiment Accuracy vs Original Accuracy with Synthetic Data Mixing
Accuracy (%)
80
60
40
20
0
Experiment Accuracy
Original Accuracy
Accuracy Gain/Loss
0% Syn 100% Real
10% Syn 90% Real
20% Syn 80% Real
30% Syn 70% Real
40% Syn 60% Real
50% Syn 50% Real
60% Syn 40% Real
70% Syn 30% Real
80% Syn 20% Real
90% Syn 10% Real
100% Syn 0% Real

Challenges

Image

1. Data Availability & Privacy
2. Data Imbalance
3. Heterogeneity of MRI Scans (scanner type,resolution,scanner thickness)
4. Noise and Artifacts (blur,scanner issues)





Streamlit deployment

https://dementiasyntheticdata.streamlit.app/

References

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[2] https://ebm.bmj.com/content/ebmed/early/2025/07/02/bmjebm-2024-113617.full.pdf

[3] https://pmc.ncbi.nlm.nih.gov/articles/PMC11851660/pdf/bioengineering-12-00186.pdf

[4] https://www.nature.com/articles/s41746-024-01409-w?utm_source=chatgpt.com

[5] https://www.mdpi.com/1424-8220/25/14/4305

[6] https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1627078/full?utm_source=chatgpt.com

[7] https://www.nature.com/articles/s41598-025-03524-4

[8] https://bmcgeriatr.biomedcentral.com/articles/10.1186/s12877-024-05026-w

[9] https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2025.1563991/full?utm_source=chatgpt.com

[10] https://pmc.ncbi.nlm.nih.gov/articles/PMC11851660/pdf/bioengineering-12-00186.pdf

THANK YOU